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Balanced basis sets of split valence, triple zeta valence and quadruple zeta valence quality for H to Rn: Design and assessment of accuracy†

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Gaussian basis sets of quadruple zeta valence quality for Rb–Rn are presented, as well as bases of split valence and triple zeta valence quality for H–Rn. The latter were obtained by (partly) modifying bases developed previously. A large set of more than 300 molecules representing (nearly) all elements—except lanthanides—in their common oxidation states was used to assess the quality of the bases all across the periodic table. Quantities investigated were atomization energies, dipole moments and structure parameters for Hartree–Fock, density functional theory and correlated methods, for which we had chosen Møller–Plesset perturbation theory as an example. Finally recommendations are given which type of basis set is used best for a certain level of theory and a desired quality of results.

Introduction

It was the first aim of the work described here to optimize high-quality Gaussian basis sets of quadruple zeta valence quality (QZV) to be used in connection with effective core potentials (ECPs) for atoms Rb to Rn—excluding the lanthanides. This was intended as a continuation of our previous definitions of segmented all-electron QZV bases for H to Kr.[1] The use of ECPs for heavier atoms is quite popular for good reasons: it reduces the number of basis functions and thus the computational work; at the same time it also permits to account for scalar relativistic effects, which improves the quality of the electronic structure calculation. Basis sets employed in conjunction with ECPs are usually of split valence (SV) or triple zeta valence (TZV) quality, commonly augmented by polarization functions. Typical examples are the LANL2DZ bases[2] on ECPs by Hay and Wadt or the Stuttgart bases for corresponding ECPs.[3–8] The TURBOMOLE SV and TZV bases[9,10] are just (slightly modified) reoptimized versions of the latter. We are not aware of consistent higher quality (than TZV) basis sets for the atoms Rb–Rn, but we note in passing, that recently basis sets of DZ, TZ and QZ quality were developed for lanthanides (ECPs from ref. 8) for calculations of bulk material.[11] Concerning elements Rb–Rn, the recent work of Peterson and coworkers[12,13] is an exception, it describes CGTOs (contracted Gaussian type orbitals) with generalized contraction schemes up to cc-pV5Z for the 4p, 5p and 6p elements (Ga–Kr, In–Xe, Tl–Rn). The work described in refs. 12 and 13 is based on recently developed ECPs[12,14] with relatively small core, *i.e.* for the *n*p elements the (n − 1)s, (n − 1)p, and the (n − 1)d shells are not included in the ECP. We have pursued the same route since it makes little sense in our opinion, to aim for high quality by using extended bases in combination with a large core ECP leaving only few electrons in the explicitly treated space. This idea was also behind the definition of the above-mentioned TURBOMOLE basis sets,[9] but at that time small-core ECPs were not available for 5p and 6p elements.

A second goal of the present work emerged during the optimization of the new QZV basis sets and the extensive tests carried out for their validation. We noticed deficiencies of the (old) TURBOMOLE bases (and of the closely related Stuttgart basis sets). These shortcomings concerned the valence basis set of the 4d and 5d elements and the polarization basis sets of these and other elements. We decided to rectify these problems and to define a large test set of more than 300 molecules for validation. The chemical compounds in this set include (nearly) all elements in their most common oxidation states and should be representative for a considerable part of chemistry. Computed molecular properties such as atomization energies, dipole moments, and structure constants then serve as a foundation to assess the accuracy that can typically be expected for the newly proposed basis sets of SV, TZV and QZV type in combination with appropriate polarization functions in Hartree–Fock (HF), density functional theory (DFT), and correlated treatments. We also identify the problematic cases for smaller basis sets. Our assessment is based on comparison of respective basis sets with QZVPP, the most extended basis set defined (here and previously[1]) by us, which in turn has been validated beforehand for typical problem cases. To the best of our knowledge this is the first thorough test of accuracies attainable for series of basis sets carried out across a large part of elements across the periodic table, involving H to Rn.

In Section 1 we present (polarized) QZV basis sets for Rb–Rn, *i.e.* considerations about numbers of primitive functions, contractions and appropriate polarization sets as well as results of representative tests. Section 2 contains the improvements made for the smaller basis sets and in Section 3 we present and discuss results arising from extensive test calculations.

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1. Basis sets of quadruple zeta valence quality for Rb–Rn

General remarks

The optimization of CGTO bases has been carried out in close analogy to our previous work concerning lighter atoms and smaller basis sets.[1,9,10,15,16] Thus only a brief characterization of procedures employed will be given. Inner shell electrons are modelled by ECPs, which reduce the required basis set size and, more importantly, account for scalar relativistic effects. The following ECPs are used: ECP-28 (core AOs 1s to 3d) for Rb[3], Sr[4], Y–Cd[5], In–Sb[14] and Te–Xe[13], ECP-46 (core AOs 1s to 4d) for Cs[3], Ba[4] and La[8], ECP-60 (core AOs 1s to 4f) for Hf–Hg[5], Tl–Bi[14] and Po–Rn[13]. As in previous work[9] these were slightly modified, since TURBOMOLE does not include g (and higher) projectors; this has a negligible effect on total energies, *e.g. ca.* 3 μE_h for Au_2, and 7 μE_h for PbO_2.

Basis set optimizations have been carried out as much as possible for the atoms at the HF level. We have minimized the HF energy with respect to orbital exponents and contractions coefficients of a segmented contraction by means of analytical gradients and a relaxation procedure, again as in the previous work.[1] This is a proven procedure, which would work virtually unaided for all-electron basis sets of comparable size. This is different in combination with ECPs, and Peterson has summarized the problems encountered here.[12]

First of all, the optimizations tend to a coalescence of adjacent orbital exponents resulting in a linearly dependent basis. This feature indicates that 3s, 4p or 5d Gaussians, which include an additional r^2, are more appropriate than the usual 1s, 2p and 3d functions.[17] Since integral and gradient codes usually can handle only the latter functions, it appears preferable to use only these. Peterson has carried out constrained optimizations with the requirement that the ratio of exponents (in the same angular symmetry) must be ≥ 1.6. This results in some loss in energy as the following example shows. If the uncontracted HF spd VDZ set for Sb[12] is subjected to our optimization procedure, the HF energy of the atomic ground state is lowered by 0.7 mE_h, the deviation to the HF limit[18] is 4.2 mE_h. (This problem has not been found for the VQZ basis of refs. 12 and 13, where analytical gradients were small and relaxation did not improve the HF energy). We have thus decided to first use the automatic optimization—until it becomes unstable as a consequence of coalescence of orbital exponents. This procedure was possible, since coalescence occurred only for steep Gaussians describing behaviour near the nucleus, which are always contracted, thus maintaining numerical stability. The corresponding two or three functions were then fixed (ratio about 1.1) and the remaining parameters fully optimized.

The spd sets

One goal of design was to keep the deviation to the HF limit, which is known for the ECPs used from numerical HF solutions,[18] well below 1 mE_h. This was achieved by the following bases, in the usual nomenclature

Rb, Sr, Cs Ba: (8s8p5d)/[7s5p3d]
Mo–Pd: (9s8p7d)/[7s5p4d]
Hf–Pt: (10s8p6d)/[7s5p4d]
In–Xe: (17s14p11d)/[7s6p4d]
Tl–At: (17s14p10d)/[7s6p4d]

For Y to Nb and for La it was sufficient to have only 8s primitives, for Ag and Cd 10s were necessary, whereas 9s sufficed for Au and Hg. This somewhat unsystematic variation of the number of primitives is at least partly due to features of ECPs, which reflect the varying degree of separation between core and outer shells. Analogous effects have been noted by Peterson and are apparent from Fig. 3 of ref. 12 which displays some scatter in the convergence of HF energies on basis set expansion.

A comparison of present ROHF energies with numerical results of Dolg[18] shows deviations for 4d and 5d elements ranging from 0.17 mE_h for Y to 0.48 mE_h for Hg. The larger errors clearly occur for the heavier elements and are mainly due to the description of d functions. This could have been easily rectified by increasing the number of d GTOs. Au is a typical example: going from 6d to 7d reduces the deviation to numerical HF from 0.38 mE_h to 0.16 mE_h. The larger d set, however, may lead to problems within a segmented quadruple zeta contractions, which requires to leave the three most diffuse functions uncontracted. In our experience this leads to a too rigid shape of the d AO, and it was decided to select the smaller 6d primitive set. As an example we consider the energetic separation of the ^2D state from the ^2S state for Au at the HF level. With the (9s8p6d)/[7s5p4d] basis this separation amounts to 70.509 mE_h, which is significantly closer to the limit,[18] 70.433 mE_h, than the value of 70.563 mE_h obtained with the (9s8p7d)/[7s5p4d] basis.

No problems of this kind occurred for the p block elements In to Xe and Tl to Rn. The differences to the HF limit thus amount to only 0.014 mE_h for Tl to 0.073 mE_h for Rn. These deviations are slightly larger than those of Peterson's VQZ basis (0.013, 0.066 mE_h). This is mainly due to using only 11d sets instead of 12d for In to Xe and 10d for Tl to At instead of 11d. It was decided to select the smaller d set, since the additional computational effort for the larger d set was not considered worthwhile. We have benefited from Peterson's work,[12,13] however. Attempts to improve the steep s functions had no success and these were adopted. We also (mainly) used his diffuse d functions serving as polarization functions. The latter was included but kept fixed in the basis set optimizations at the HF level.

The 8p primitive set for Rb to Hg includes two diffuse functions to describe the 5p and 6p AO, respectively, which serve as polarization functions. These were optimized in trial HF calculations for the ^2P states of Rb, Cs, and the ^3P states of Sr, Ba, Cd and Hg. For the elements in-between orbital exponents were obtained by interpolation. The remaining six p functions describing the occupied 4p (Rb–Cd) and 5p AOs (Cs–Hg) were then optimized by keeping the two diffuse functions fixed. Only the four steepest p functions were contracted to guarantee an accurate description of the ^2S–^2P separation in Rb, Cs, and ^1S–^3P splitting in Sr, Ba, Cd and Hg. The d functions for Rb, Sr, Cs, Ba have finally been optimized in HF treatments of ^2D and ^3D states, with remaining basis functions kept as determined beforehand.

Higher polarization functions

It is reasonable to distinguish between HF and DFT treatments on one hand and correlated calculations on the other hand. We thus propose two sets of polarization functions denoted 'P' and 'PP', as described in Table 1. The procedure is again analogous to that described previously[1] for the 4p elements. The PP polarization basis is designed for correlated treatments, for DFT the slightly smaller P sets are considered sufficient. P and PP are identical for the p elements. In Table 1 we further distinguish between core and valence polarization to indicate their purpose. The core functions refer to the occupied d shell below the valence shell for the p elements, and to the s and p core orbitals below the valence s shell of Rb, Sr, Cs, and Ba. The valence PP functions for the p elements have been taken from Peterson.[12,13] The other functions in P and PP have been optimized in atomic MP2 calculations for representative elements and interpolated in-between. The number of core functions, Table 1, has been set after numerous tests described in more detail in the subsequent subsection.

Table 1 Specification of QZV, QZVP and QZVPP basis sets for representative elements. QZVP basis sets include all functions of columns 'QZV' and 'P', QZVPP basis sets those of columns 'QZV' and 'PP'. 'Valence' denotes functions for polarization of valence shells, 'Core' steeper polarization functions to polarize the inner shells. Numbers of electrons included in the effective core potential are listed in column 'ECP'

		QZV	P		PP	
Atom	ECP	Valence	Core	Valence	Core	Valence
Sb	28	(17,14,11)/[7,6,4]	2f	2f1g	2f	2f1g
Bi	60	(17,14,10)/[7,6,4]	2f	2f1g	2f	2f1g
Tc	28	(9,8,7)/[7,5,4]		3f1g		4f2g
Re	60	(10,8,6)/[7,5,4]		3f1g		4f2g
Rb, Sr	28	(8,8,5)/[7,5,4]	1f		2f	1f
Cs	60	(8,7,5)/[6,5,3]	1f		2f	1f
Ba	60	(8,8,5)/[7,5,3]	1f		2f	1f

Representative tests

As in the preceding work[1] results obtained with the QZVPP bases at levels HF, DFT(BP86)[19–21] and MP2[22] were compared to those obtained with larger basis sets; we investigated atomization energies for At_2, SbF_5 and SbF_3 as compounds containing p elements only, Au_2, AgCl and ZrO_2 as representatives for compounds containing d and p elements, and BaF_2 as a typical (ionic) combination of s and p elements. Preliminary calculations indicated these to be critical examples.

We required the energy to be converged within 10^{-6} E_H for HF and DFT, in case of MP2 additionally the norm of the (HF) density within 10^{-7}. In case of DFT the grid for numerical evaluation of the exchange–correlation operator was of medium size (gridsize 3).[21] At the MP2 level the inner shells were excluded from correlation treatment: 1s for O and F, 1s–2p for Cl and 5s5p (6s6p) for Sb (At); for the other cases all electrons not included in the ECP were correlated. Structure parameters for all compounds resulted from optimizations at the DFT(BP86)/SV(P) level. As reference bases we used for O and F cc-pV5Z bases,[23] [6s5p4d3f2g1h], for Cl cc-pV(5 + d)Z,[24] [7s6p5d3f2g1h], for At and Sb cc-pV5Z-PP[12,13], which we extended by two additional steep f functions, leading to [7s7p5d5f2g1h]. For Au, Ag, Zr and Ba we (partially) decontracted and additionally polarized QZVPP bases yielding the following contraction schemes:

Au, Ag: {111111111/2111111/21111/111111/1111/11}, [9s7p5d6f4g2h]

Zr: {11111111/311111/31111/111111/1111/11}, [8s6p5d6f4g2h]

Ba: {11111111/311111/2111/11111/11/1}, [8s6p4d5f2g1h].

Results for bond energies per atom are collected in Table 2. At levels HF and DFT we find that the differences between QZVPP and the reference bases typically amount to less than 0.01 eV ($\approx$1 kJ/mol), for the ionic compounds ZrO_2 and BaF_2 they are slightly larger, *ca.* 0.017 eV. At the MP2 level the differences are larger, typically up to 0.04 eV, but amount to 0.072 eV (0.062 eV) for BaF_2 (ZrO_2) and even 0.14 eV for Au_2. The latter may be regarded as an extreme case, as electron correlation is the dominant part for the bond (*ca.* 70% at the MP2 level, see Table 2), which is of high complexity, as the d and s shells are nearly degenerate for Au. The deviation just documents the slow convergence of correlation energies on basis set extension. The errors for these compounds are apparently larger than those for analogous compounds of elements up to Kr presented previously.[1] This is mainly due to the choice of larger reference bases for Ba and the transition metals. Further note that, in contrast to HF and DFT, for MP2 the basis set limit is not yet reached with the reference bases, which means that in fact the differences to the MP2 basis set limit are even somewhat larger. In praxis, it is usually not reasonable to perform MP2 calculations with such large basis sets, as the accuracy of the method is not worth that effort, but the above errors for MP2 may serve as a guide for other more sophisticated (and much more costly) post-HF methods, like coupled cluster or CI.

For At_2, SbF_5 and SbF_3 the results obtained with the QZVPP bases are much closer to the reference than those obtained with cc-pVQZ-P and cc-pV5Z-PP bases[12,13] at levels DFT and HF. These errors can directly be traced back to the lack of steep f functions necessary for the polarization of the inner d shell for these bases, which is the only difference between the reference basis and cc-pV5Z-PP for At and Sb. Similarly it was found previously[24] that for Al–Ar a steep d function to polarize the inner p shell improves results in the same way.

At the MP2 level QZVPP underestimates the bond energies for the p element compounds (as well as for the others), whereas the larger cc-pV5Z-PP basis always yields larger bond energies than the reference bases. This results from the inclusion of the d shells in the correlation treatment in combination with the lack of steep f functions. In case of the cc-pVQZ-PP basis this partly compensates the usual underestimation of bond energies typical of smaller basis sets, thus leading to apparently better results.

2. Basis sets of split valence (SV) and triple zeta valence (TZV) quality for H–Rn

In previous works[1,9,10,15,16] basis sets of SV and TZV quality plus smaller (P) and larger (PP) sets of polarization functions

Table 2 Bond energies per atom, in eV, at levels DFT(BP86), HF and MP2 for reference basis sets and errors for smaller bases. For details concerning the reference bases and the frozen core approximation (MP2) see text

		DFT	HF	MP2
At_2	Reference	0.978 058	0.226 708	1.047 582
	Δ(def2-QZVPP)	−0.001 034	−0.000 394	−0.026 408
	Δ(cc-pVQZ-P)	−0.005 850	−0.005 129	−0.012 353
	Δ(cc-pV5Z-P)	−0.004 272	−0.004 163	+0.037 605
SbF_5	Reference	3.445 021	1.886 285	3.525 779
	Δ(def2-QZVPP)	−0.004 712	−0.006 340	−0.031 124
	Δ(cc-pVQZ-P)	−0.012 190	−0.023 995	+0.027 741
	Δ(cc-pV5Z-P)	−0.023 324	−0.018 893	+0.052 712
SbF_3	Reference	3.624 239	2.014 205	3.635 260
	Δ(def2-QZVPP)	−0.004 027	−0.005 768	−0.034 782
	Δ(cc-pVQZ-P)	−0.012 673	−0.024 224	−0.016 598
	Δ(cc-pV5Z-P)	−0.018 646	−0.017 585	+0.021 911
ZrO_2	Reference	5.174 425	2.685 607	5.392 149
	Δ(def2-QZVPP)	−0.017 396	−0.018 911	−0.062 721
AgCl	Reference	1.557 530	1.106 018	1.678 837
	Δ(def2-QZVPP)	−0.008 258	−0.007 211	−0.029 034
Au_2	Reference	1.130 236	0.399 756	1.278 889
	Δ(def2-QZVPP)	−0.004 639	−0.003 891	−0.013 959
BaF_2	Reference	4.103 600	2.841 300	4.196 471
	Δ(def2-QZVPP)	−0.015 963	−0.020 000	−0.072 027

Table 3 Numbers of contracted basis functions for all TURBOMOLE (default-) basis sets for representative elements. The notation '4321', *e.g.*, is an abbreviation for [4s3p2d1f] and denotes a (contracted) basis of type 4s3p2d1f

	def2-						def-			
	QZVPP	QZVP	TZVPP	TZVP	SVP	SV(P)	TZVPP	TZVP	SVP	SV(P)
H	4321	4321	321	31	21	2	321	31	21	2
N	74321	74321	5321	5321	321	321	5321	531	321	321
P	96421	96421	5531	5521	431	431	5521	541	431	431
As	(11)7441	(11)7421	6541	6541	543	543	6541	653	543	543
Sb	76441	76441	6532	6532	442	442	2321	231	221	221
Bi	76441	76441	6532	6532	442	442	3321	331	221	221
Mn	(11)6542	(11)6531	65421	6441	5321	532	6431	643	532	532
Tc	75442	75431	64321	6431	5321	532	5331	533	532	532
Re	75442	75431	64321	6431	6321	632	6331	633	632	632
Li	6421	6421	531	53	32	32	53	51	31	31
Na	9542	9531	543	543	421	421	541	53	42	42
K	(11)643	(11)641	643	643	532	532	651	64	53	53
Rb	7543	7541	6431	643	532	532	531	53	53	53
Cs	6533	6531	5331	533	532	532	531	53	53	53
Be	7421	7421	531	531	32	32	53	52	32	32
Mg	9552	9541	543	543	431	431	541	53	42	42
Ca	(11)643	(11)641	653	653	532	532	653	642	532	532
Sr	7543	7541	6431	643	432	432	432	432	432	432
Ba	7533	7531	6431	6431	432	432	432	432	432	432

have been presented for elements H–At (except for the lanthanides) and their accuracy was shown for selected examples. Due to the nomenclature in the TURBOMOLE basis set library, these bases are termed 'def-bases' ('def' is an abbreviation for 'default'). In this work we present an improved, (partly) new series of such bases, termed 'def2', and carry out extended studies to assess their quality. A short description by contraction patterns is given in Table 3, further details are discussed below. The polarized QZV bases presented above and in the preceding work[1] are denoted def2-QZVP and def2-QZVPP from this point on, to indicate that they consistently refer to small-core ECPs.

TZV and SV bases for In–Xe and Tl–Rn

For these elements bases specified in our previous works were designed to be used with comparably large ECPs covering the inner 46 (78) electrons, whereas the QZV bases described above fit to smaller ECPs covering 28 (60) electrons. We decided to develop new basis sets of TZ and SV quality for the smaller ECPs. These new TZV bases have [6s5p3d] CGTOs throughout, see Table 3, *i.e.* for each angular momentum just one contraction less than QZV described above. The number of primitives was chosen such that the deviation to the HF limit is below 1 mE_h, which resulted in (10s9p8d) for In–Sb and Tl–Bi, (11s9p8d) for Te and (11s10p8d) for the remaining heavier elements. These basis sets include a diffuse d function to polarize the valence shell, which was kept fixed but included in the optimization of the remaining s, p and d functions. The f function was taken from the literature.[12,13]

The SV bases were of (10s7p6d)/[4s4p2d] type throughout. The d set includes a polarization function as for TZV. Deviations of atomic HF energies from the limit range from 1.5 mE_h for Tl to 6.6 mE_h for I. Since the rare gases Xe and Rn are special cases for which using an SV basis appears not recommendable, we have equated SV and TZV for these elements.

TZV bases for 4d and 5d elements

In the course of systematic basis set optimizations carried out within this work, it was decided to define also new triple zeta bases for the 4d and 5d elements (employing the same ECPs as for QZV and as in the previous work[1]). The previous TZV had sizes (7s6p5d)/[5s3p3d] for 4d and (7s6p5d)/[6s3p3d] for 5d elements, see Table 3. For Y–Rh we have only added one p set and decontracted the s yielding (7s7p5d)/[6s4p3d]. This lowered the atomic energies by up to 26.7 mE_h and reduced the deviation from the HF limit to 0.7 mE_h in case of Rh. For other elements we also added primitives to get the energy within about 1 mE_h of the HF limit, the number of contractions is [6s4p3d] for all 4d and 5d elements.

Polarized bases of SV and TZV quality for H–Rn

At the TZV level we propose two different sets of polarization functions, denoted P and PP. def2-TZVPP, specified in Table 3, includes the standard polarization set for H (2p1d)[23] and p elements up to Kr, which is (2d1f)[23,25] plus an additional steep d function for Al–Ar.[24] For 5p and 6p elements it was found necessary to have 2f valence functions instead. A (2f1g) polarization basis is used for all d elements. The s elements contain polarization sets from [3p1d] for Li to [3d1f] for Sr and Ba. The polarization sets are identical to those of the def-TZVPP bases for H, B–Ne and Ga–Kr, but are larger for s and d elements; these extensions emerged after numerous tests for the sample of more than 300 molecules considered below. These tests also showed that slightly fewer polarization functions suffice for many purposes, especially for DFT treatments. The def2-TZVP bases thus include only (1p) for H, (1f) for d elements and partly reduced sets for s elements. For p elements def2-TZVP bases are usually identical to def2-TZVPP, but for Al–Ar the additional steep d function was contracted with the steeper of the remaining d functions yielding a (3d1f)/[2d1f] polarization set.

Bases of type def2-SV(P) contain a polarizing d function for p elements and a diffuse p set for d elements, but no polarization functions for H, just as def-SV(P). For the s elements comparably large polarization set are included in def2-SV(P), see Table 3. It turned out that sometimes a polarizing p function for H and an f function for the d elements is necessary; these additional functions are included in the def2-SVP bases.

Comparison with small-core basis sets for 5p and 6p elements from the literature

Peterson *et al.* have recently presented cc-pVXZ-PP (X = D, T, Q) basis sets[12,13] for 5p and 6p elements and we will briefly compare these with the present bases. The main difference concerns the contraction pattern. Peterson *et al.* employ a generalized contraction of optimized atomic HF basis sets (for occupied AOs). The present bases use segmented contractions with orbital exponents and contraction coefficients optimized for atomic HF ground states. This is consistent with the other TURBOMOLE basis sets and is much more efficient for programs not designed for generalized contractions.[1] The present basis sets also include steep polarization functions for the triple zeta and the quadruple zeta basis sets, which are lacking in refs. 12 and 13. The importance of these functions has been discussed above (Section 1, representative tests) as well as in ref. 1. As a consequence we get slightly improved atomization energies at DFT and MP2 level, as pointed out in Section 1 for quadruple zeta bases and similarly for the triple zeta bases. Only insignificant differences were found for split valence bases.

3. Results of molecular tests

General considerations

For all def- and def2-bases test calculations were carried out for a large test set of more than 300 compounds[26] representing (nearly) each element in (nearly) each common oxidation state. Atomization energies at levels HF, MP2 and DFT(BP86) and dipole moments at levels HF and DFT(BP86) were calculated for all compounds in the DFT(BP86)/SV(P) geometry (available as supplementary material). We define some terms that are needed for the discussion of results. For a basis X, size(X) denotes the ratio of basis functions of X to that of def2-QZVPP taken over the entire test set. Of course, this ratio depends on the test set, but as the present choice of molecules is quite representative for usual applications, it is a reasonable guide. The quality of a basis set will be characterized by the difference of results to those obtained with def2-QZVPP bases for the entire test set, in the following termed 'errors'; the results will be characterized by mean errors and their standard deviation. High quality means both small mean errors and small standard deviation of errors and also a small number of molecules with atypically large errors.

It was found convenient to partition the test set[26] into three groups of compounds: molecules that contain only hydrogen and elements from groups 13 to 18, in the following denoted 'p compounds', those that also contain transition metals, 'd compounds', and those that also contain (earth) alkaline metals (but no transition metals), 's compounds'. A similar quality for these three classes of compounds is desirable for every type of basis corresponding to a balanced description all across the periodic table. Moreover, the quality of results should increase steadily with the basis set size.

Main results

Results referring to the whole test set are sampled in Table 4. For split valence and triple zeta bases these results are also graphically shown in Figs. 1–5 together with the respective data resulting from a partitioning of the test set in s, p and d element compounds. Let us first of all note the main results. For the def2-series one observes a consistently increasing quality with increasing basis set size, Table 4, for all quantum chemical methods tested. As a typical example, standard deviations of errors in atomization energies per atom at the DFT level amount to *ca.* 0.1 eV for the smallest bases, def2-SV(P), they are somewhat reduced for def2-SVP, much smaller for def2-TZVP (0.022 eV), which is similar to def2-TZVPP (0.018 eV), and again smaller for def2-QZVP (0.004 eV). Similar trends are observed for the mean errors and for any other method and property tested. This is a clear improvement compared to the previous def-bases. Moreover, for all quantum chemical methods tested here, def2-bases provide a well balanced description all over the periodic table for all types of bases. Nevertheless, there are subtle noteworthy differences. At the split valence level, *e.g.*, def2-SV(P) yields reasonably balanced errors for DFT, whereas for MP2 (and also for HF) def2-SVP bases are necessary to keep the errors of d compounds in the range of s or p compounds.

Let us discuss the results for the def2-bases in detail to prepare for recommendations which type of basis is to be used best for a given method for a desired quality of results.

Atomization energies

We start with the results for bond energies, results for dipole moments and for structure parameters are discussed afterwards. For def2-SV(P) bases, which differ from the previous def-SV(P) bases only in the polarization of s elements, mean errors and standard deviations at the DFT level are similar for

Table 4 Basis set size and accuracy of def- and def2-basis sets compared to def2-QZVPP. Mean values, ϕ, and standard deviations, σ, result from calculations of atomization energies at levels HF, DFT(BP-86) and MP2[a] and dipole moments at levels DFT(BP-86) and HF for a set of 311 representative molecules

	def2-					def-			
	QZVP	TZVPP	TZVP	SVP	SV(P)	TZVPP	TZVP	SVP	SV(P)
Basis set size compared to def2-QZVPP[b]									
AO[c]	0.954	0.564	0.489	0.268	0.244	0.499	0.318	0.248	0.238
CAO[c]	0.944	0.527	0.447	0.233	0.209	0.456	0.269	0.212	0.203
Differences to def2-QZVPP in atomization energies[d]/meV (311 molecules)									
DFT ϕ	−1	−11	−27	−21	−60	−38	−113	−68	−80
σ	4	18	22	91	102	67	111	140	139
HF ϕ	−2	−21	−38	−92	−150	−65	−166	−151	−167
σ	6	23	35	108	159	78	158	177	168
MP2 ϕ	−3	−96	−119	−312	−387	−115	−338	−362	−411
σ	14	43	52	156	163	92	163	202	176
Differences to def2-QZVPP in (non-zero) dipole moments/mD (166 molecules)									
DFT ϕ	7	0	15	−135	−115	65	179	−112	−102
σ	22	28	75	295	311	190	294	371	382
HF ϕ	10	28	68	22	40	122	272	117	124
σ	25	74	142	258	323	262	419	454	459

[a] Frozen core orbitals: B–Mg: 1s, Al–Zn: 1s2s2p, Ga–Kr: 1s2s2p3s3p, In–Xe: 4s4p, Tl–Rn: 5s5p. [b] Average value for all molecules in test set. [c] AO: spherical harmonic basis functions (1s3p5d7f...), CAO: cartesian basis functions (1s3p6d10f...). [d] For anions the difference to the neutral electrons is calculated, *i.e.* the sum of atomization energy and vertical detachment energy.

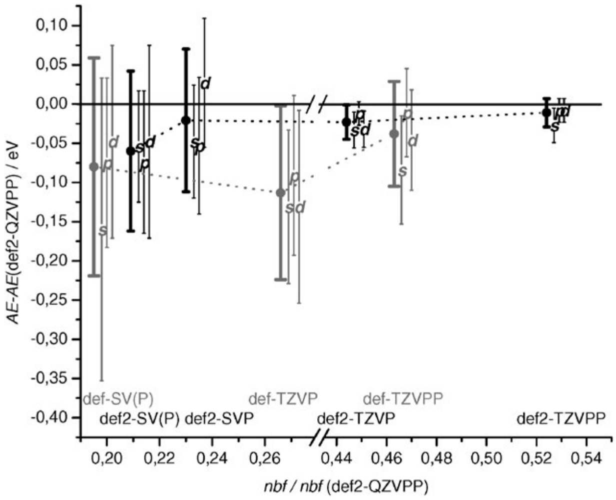

Fig. 1 Deviations in atomization energies per atom, AE, in eV, at DFT(BP86) level for basis sets of split valence and triple zeta valence quality to results obtained with def2-QZVPP bases ('errors'). Data of bases presented in previous works are printed in gray, such of bases presented in this work are printed in black. By filled circles the average errors for the whole test set are displayed, the bars indicate the standard deviation of errors. 'p' denotes the respective values for a subset containing only compounds consisting of p elements (and H), 's' that for compounds that additionally consist of s elements, and 'd' that for compounds that additionally consist of d elements (but not of s elements). The abscissa position for each basis set type is given by the ratio of the number of basis function used for the whole test set with the respective basis and that of def2-QZVPP.

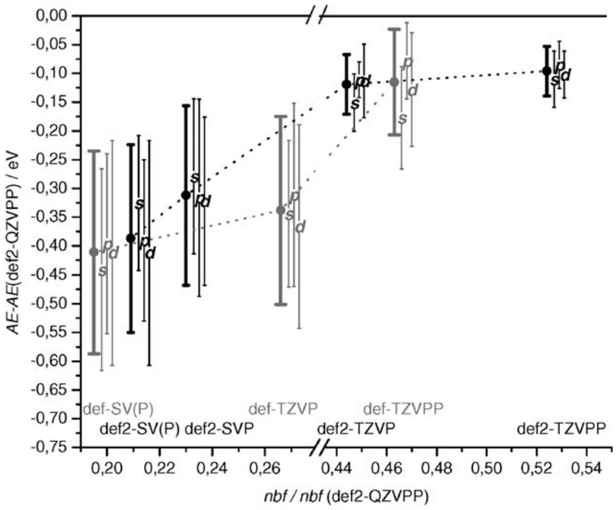

Fig. 3 Errors in atomization energies at the MP2 level. For explanations see Fig. 1.

all types of compounds amounting to *ca.* 0.1 eV atom^{-1}, which is not much larger than the typical accuracy to be expected from current DFT methods. The largest errors in atomization energies (per atom) are observed for ionic compounds (KF: −0.26 eV), extreme oxidation states (OsO_4: −0.36 eV) and for anions (IO_4^-: −0.47 eV); for the latter the value is the difference between the anionic compound and the (neutral) atoms, *i.e.* the sum of atomization energy and vertical detachment energy. For the def2-SVP bases d compounds are described better than the others at the DFT level, as the additional f function (which was added to improve HF-results at the split valence level) often significantly improves the energies of molecules, but not that of atoms. Due to this effect the mean error of def2-SVP bases at the DFT level is very small, and the standard deviation is slightly reduced compared to def2-SV(P) (Table 4). But, as mean errors for different types of compounds are quite different for DFT/def2-SVP, this combination should be used with care.

A balanced description all across the periodic table for HF and MP2 at split valence level is obtained rather with the def2-SVP bases, which thus are recommended to be used for these procedures as small bases to get qualitatively reasonable results. Errors for HF/def2-SVP are in the same range as for DFT/def2-SV(P), worst cases are IO_4^- (−0.52 eV) and BaO (−0.44 eV). Errors for MP2/def2-SVP are larger due to the well-known slower basis set convergence for post-HF methods compared to HF or DFT. Atomization energies per atom are typically 0.3 eV too small and show a standard deviation of *ca.* 0.15 eV, but in critical cases errors may become much larger (BaO: −0.7 eV, IO_4^-: −1.02 eV, MnO_4^-: −1.04 eV).

At the DFT/def2-TZVP level mean errors and standard deviations are very small (*ca.* 0.02 eV atom^{-1}, which usually is below the error of current DFT methods) and very similar for all types of compounds. Largest errors occur either for ionic compounds (BaO, RhF_6: −0.09 eV atom^{-1}) or for rare gas compounds ($XeOF_4$: −0.07 eV atom^{-1}). Thus the def2-TZVP basis is suited for quantitatively accurate DFT treatments. Despite the fact that this basis is smaller than the (old) def-TZVPP basis, the quality is very much improved by the modifications described above. For similar accuracy (0.02 eV atom^{-1} and balanced errors for all types of compounds) within

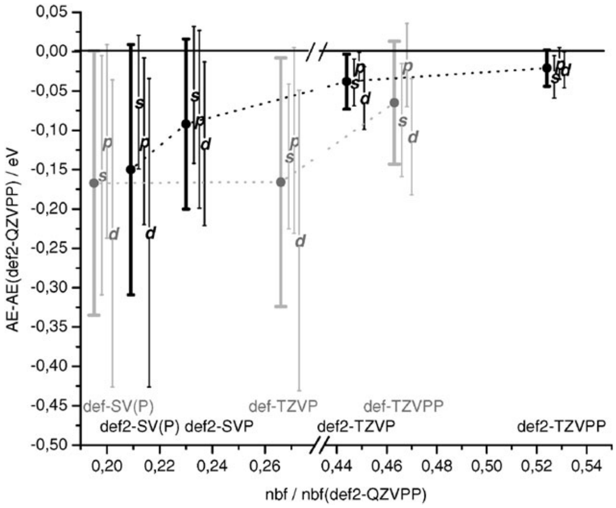

Fig. 2 Errors in atomization energies at the HF level. For explanation see also Fig. 1.

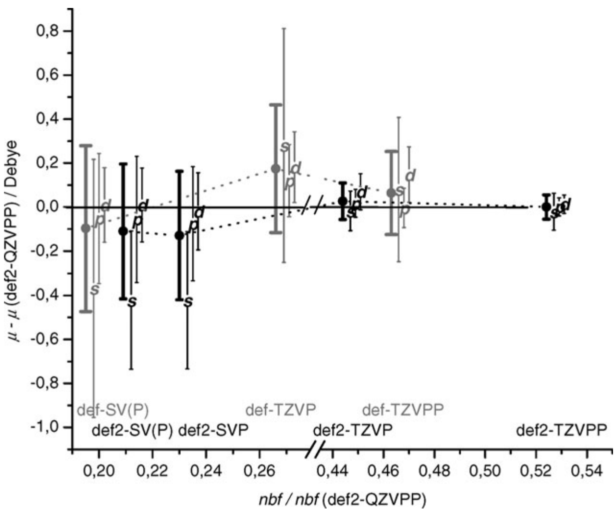

Fig. 4 Errors in dipole moments, μ, in D, at the DFT(BP86) level for the 166 systems with non-vanishing dipole moments of the molecular test set. For explanations see Fig. 1.

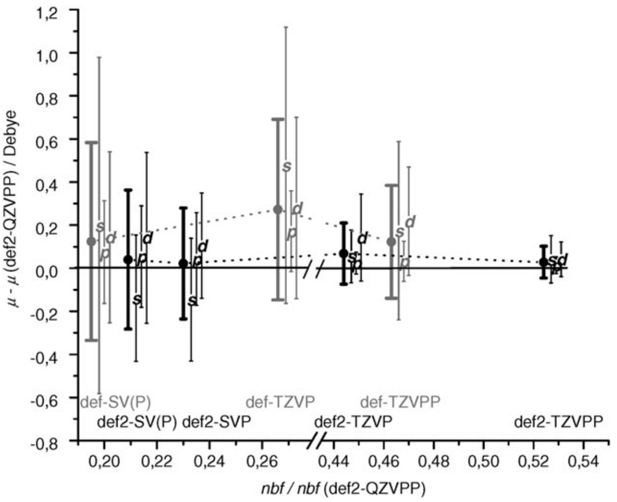

Fig. 5 Errors in dipole moments, μ, in D, at the HF level for the 166 systems with non- vanishing dipole moments of the molecular test set. For explanations see Fig. 1.

HF, def2-TZVPP bases (larger polarization set than def2-TZVP for s and d elements) have to be used; worst cases are similar as above CoF_2 (−0.17 eV), BaO (−0.13 eV), and $XeOF_4$ (−0.11 eV). For MP2 treatments at the triple zeta basis set level def2-TZVPP bases are also a better choice than def2-TZVP. For this method the errors per atom, *i.e.* the differences to the def2-QZVPP results, are also balanced for different types of compounds, but significantly larger than for HF or DFT (mean value *ca.* 0.1 eV, standard deviation *ca.* 0.05 eV, the largest error of 0.21 eV occurs for MnO_4^-, IO_4^- and CsF). Regarding the fact that for MP2 (and any other post-HF method) the basis set limit is not reached with quadruple zeta bases, the differences to this limit are even larger. Nevertheless, def2-TZVPP bases provide a balanced description at MP2 level both in the sense of consistent errors all across the periodic table and in the sense of similar errors for the basis set and the method itself.

Dipole moments

For the 166 systems with non-vanishing dipole moments we checked that for any molecule the vector of the dipole moment pointed into the same direction for all bases and thus just regarded the absolute values. Statistical evaluations (for this subset of molecules) for levels HF and DFT are given in the lower part of Table 4, graphical representations also including the data resulting from a split into s, p and d compounds are shown in Figs. 4 and 5. As noted above, for the def2-bases we observe a consistent improvement of quality with increasing basis set size for both DFT and HF, we further find an improvement over the corresponding (old) def-bases. Regarding Figs. 4 and 5, it is most striking that for the split valence bases the mean error of s compounds is significantly larger than for p or d compounds (whereas standard deviations are similar) for both DFT and HF. These larger absolute errors (relative errors are clearly smaller) occur due to the ionic character of most of the s compounds, which leads to problems in the description of the anionic part of the compound, *i.e.* the p element and not the s element, with a comparably small basis. This is a general problem for the description of anionic compounds with small bases and not a problem of the def2-SV(P) or def2-SVP bases for the *s* elements.

In case of DFT, the improvement of def2-SVP *vs.* def2-SV(P) indeed is not significant, in both cases standard deviations are *ca.* 0.3 D, the absolute value of the mean deviation even slightly increases from def2-SV(P) to def2-SVP. For def2-TZVP bases errors are typically below 0.1 D for all groups of compounds, and thus in the range of errors of current DFT methods. Nevertheless for DFT dipole moments at the triple zeta level def2-TZVPP bases provide a significant improvement over def2-TZVP; standard deviations are much smaller and amount to 0.028 D. For the whole test set, independent of the basis, largest errors in both directions (over- and underestimation of the absolute values) are observed always for compounds containing elements O, F, S or Cl. In case of def2-SV(P) bases by far worst is K_2S (−1.17) followed by TlCl (−0.67) and H_3PO_4 (+0.62), for def2-TZVP we get ScO (+0.35), CsO (−0.18) and H_2O (+0.18). For the def2-QZVP basis differences to def2-QZVPP are very small, even for the worst cases they are always below 0.1 D.

At the HF level it is advisable to use the higher polarized bases def2-SVP and def2-TZVPP, as both standard deviation and mean errors are significantly reduced compared to the corresponding lower polarized bases def2-SV(P) and def2-TZVP. HF dipole moments are typically somewhat too large as compared to def2-QZVPP and show standard deviations of *ca.* 0.3 D (def2-SVP) and 0.074 D (def2-TZVPP). With the def2-SVP bases we observe largest underestimations of absolute values of HF dipole moments for sulfides (BaS: −0.80 D, Al_2S_3: −0.63 D, NiS: −0.74 D), largest overestimations for O or F containing compounds (WO_2: +0.61 D, H_3PO_4: +0.46 D, BaF: +0.39 D). For the def2-TZVPP bases trends are less clear, worst cases are K_3P (+0.35 D), RuF (+0.22 D), NiS (−0.15 D) and Na_3P (−0.14 D).

Equilibrium geometries

Optimized structure parameters resulting from analytical gradients of the energy followed by a relaxation procedure were calculated at the DFT level for all compounds of the test set using bases def2-QZVPP, def2-TZVP and def2-SV(P). The resulting large number of data was reduced to a few quantities by regarding only bond distances, which were defined as distances between two atoms that are smaller than three times the sum of Bragg–Slater radii. All bond distances obtained with def2-SV(P) and def2-TZVP bases were compared to those obtained with def2-QZVPP bases. We formed the mean value of differences, $\phi(d)$, as well as that of absolute values of differences, $\phi(|d|)$. For the def2-SV(P) bases we find $\phi(d)$ = 1.32 pm and $\phi(|d|)$ = 1.63 pm, *i.e.* equilibrium distances with this basis are typically 1–2 pm too large compared to the basis set limit. Errors are similar for s, p and d compounds, which means that for DFT structure optimizations usually def2-SV(P) yields reasonable results. Nevertheless, for problematic cases as *e.g.* ionic compounds or weakly bound systems errors may be larger; worst cases are BaO, CsO (+*ca.* 8 pm) and Li_2(+6.1 pm) among the s compounds, Cl_2, S_5 and ICl (+*ca.* 5 pm) among the p compounds, and AuCl, $AuCl_3$, Au_2 and Au_3^- (+*ca.* 5 pm) among the d compounds. For the def2-TZVP bases one gets $\phi(d)$ = 0.23 pm and $\phi(|d|)$ = 0.32 pm. For s compounds the errors are larger than for the others, $\phi(d)$ = 0.48 pm and $\phi(|d|)$ = 0.78 pm. Errors larger than 1 pm were found for RbO, CsO, KF (+*ca.* 2 pm) and 17 other s compounds that show errors up to 1.75 pm, Au_3^-, Au_2, Cu_2, Ag_2 (+1.3…+1.8 pm) and AuCl(+1.1 pm). For the p compounds largest errors are observed for $XeOF_4$(+0.78 pm).

Next, HF and MP2 structure optimizations were carried out for a few selected compounds that turned out to be problematic in at least one of the previous investigations (and additionally for water) leading to the results collected in Table 5. In case of MP2 numerical gradients were used, as in TURBOMOLE the frozen core approximation for gradients of the MP2 energy is implemented only in connection with the RI-approximation, which would require optimized auxiliary basis set that are not yet available for all elements and (orbital) bases.

For HF matters are similar as for DFT, typically errors for def2-TZVPP bases (*i.e.* differences to def2-QZVPP) are clearly

Table 5 Structure parameters (bond lengths/pm, angles/°) for selected compounds at levels DFT(BP86), HF and MP2 (for details of the frozen core approximation see Table 4) with all def2-bases

		QZVPP	QZVP	TZVPP	TZVP	SVP	SV(P)
DFT							
H_2O	$d_{O\text{–}H}$	96.91	(=QZVPP)	96.95	97.13	97.51	98.12
	$\angle_{H\text{–}O\text{–}H}$	104.14	(=QZVPP)	103.96	104.36	102.17	103.53
PbO_2	$d_{Pb\text{–}O}$	189.63	(=QZVPP)	189.90	(=TZVPP)	193.32	(=SVP)
Au_2	$d_{Au\text{–}Au}$	251.31	251.73	251.86	252.80	254.37	256.47
Cu_2O	$d_{O\text{–}Cu}$	177.21	177.23	177.24	177.51	177.26	177.41
	$\angle_{Cu\text{–}O\text{–}Cu}$	89.99	90.10	90.29	89.89	86.22	86.10
Mg_4	$d_{Mg\text{–}Mg}$	309.18	309.23	310.02	(=TZVPP)	310.51	(=SVP)
BaO	$d_{Ba\text{–}O}$	199.13	199.42	200.49	(=TZVPP)	207.56	(=SVP)
HF							
H_2O	$d_{O\text{–}H}$	93.96	(=QZVPP)	94.00	94.19	94.49	95.03
	$\angle_{H\text{–}O\text{–}H}$	106.34	(=QZVPP)	106.26	106.52	105.14	106.36
PbO_2	$d_{Pb\text{–}O}$	184.63	(=QZVPP)	184.79	(=TZVPP)	188.92	(=SVP)
Au_2	$d_{Au\text{–}Au}$	260.21	260.50	260.32	260.72	262.07	263.15
Cu_2O	$d_{O\text{–}Cu}$	180.71	180.66	180.74	180.88	180.03	180.23
	$\angle_{Cu\text{–}O\text{–}Cu}$	151.81	151.80	141.20	132.35	120.63	119.29
BaO	$d_{Ba\text{–}O}$	194.01	194.01	195.01	(=TZVPP)	204.22	(=SVP)
MP2							
H_2O	$d_{O\text{–}H}$	95.80	(=QZVPP)	95.86	96.22	96.25	96.96
	$\angle_{H\text{–}O\text{–}H}$	104.22	(=QZVPP)	103.82	104.60	102.43	105.23
PbO_2	$d_{Pb\text{–}O}$	187.34	(=QZVPP)	187.84	(=TZVPP)	190.45	(=SVP)
Au_2	$d_{Au\text{–}Au}$	242.72	244.11	243.72	245.78	248.89	257.30
Cu_2O	$d_{O\text{–}Cu}$	172.09	172.36	173.84	173.78	177.66	179.02
	$\angle_{Cu\text{–}O\text{–}Cu}$	93.06	94.31	93.37	94.27	88.46	88.79
Mg_4	$d_{Mg\text{–}Mg}$	300.26	300.25	304.88	(=TZVPP)	309.35	(=SVP)
BaO	$d_{Ba\text{–}O}$	198.39	198.53	199.77	(=TZVPP)	210.95	(=SVP)

below 1 pm and even for def2-SV(P) they are not much larger. At the HF level Mg_4 was omitted, as it is bound mainly by effects of electron correlation. BaO, which previously turned out to be a difficult case (DFT and HF bond energies, DFT structure parameters), shows larger errors, 1 pm for def2-TZVPP and even 10 pm for the def2-SV(P) basis. This error can be traced back to deficiencies in the polarization set of the def2-SV(P) basis for Ba. If one uses the 3d1f set from the def2-TZVP basis instead of the original 2d set, the error decreases to 0.5 pm. Nevertheless, such a large polarization set does not really fit the description of the valence shell by only two (contracted) sets. The bond angle of Cu–O–Cu shows an atypical, very pronounced basis set dependence: it amounts to *ca.* 120° for def2-SVP, 141° for def2-TZVPP and 152° for def2-QZVPP. This system is very sensitive, as it is right in-between a purely ionic compound with a linear structure and a three center bond leading to a bent one. Matters are less critical for the less ionic compound Cu_2Se. The values obtained with def2-(QZVPP/TZVPP/SVP) bases for the bond angle are (97.99°/96.14°/89.06°). We summarize, that def2-SVP bases usually are sufficient for qualitatively correct results at the HF level and that with def2-TZVPP bases errors in bond lengths typically are smaller than 1 pm and that in bond angles smaller than 1°.

Errors at the MP2 level are larger than at the DFT or at the HF level, as expected from the slower basis set convergence for this method. In case of the def2-TZVPP basis errors for Au_2, Cu_2O and BaO amount to 1–2 pm, for Mg_4, which is (very weakly) bound solely by correlation effects, they are somewhat larger (4.5 pm), whereas for the less critical p compounds PbO_2 and H_2O they are clearly below 1 pm. With the def2-SVP basis we get errors of +1 (+3) pm for H_2O (PbO_2) and errors of +5 to +12 pm for the s and d compounds. In case of Au_2 the polarizing f function is of great importance; if it is omitted, as in def2-SV(P), the Au–Au distance increases by further 8.5 pm. A missing f function also is the main reason for the large error (+13 pm) in case of BaO for def2-SVP, but for reasons discussed above, we decided not to add this function for the split valence basis. The bond angle of Cu_2O at the MP2 level (as well as at the DFT level) is much less sensitive to basis set changes than for HF, probably because electron correlation favors the three-center *vs.* the ionic bond.

Summary

We have presented Gaussian basis sets of quadruple zeta valence quality for Rb–Rn, denoted def2-QZVP and def2-QZVPP, as well as bases of split valence and triple zeta valence quality for H–Rn, def2-SV(P), def2-SVP, def2-TZVP and def2-TZVPP. The latter were obtained by (partly) modifying previously developed bases def-SVP, def-SV(P), def-TZVP and def-TZVPP. A large set of molecules representing (nearly) each element in all its common oxidation states was used to test and to assess the quality of def- and def2-bases.

As a summary we propose recommendations for the use of the basis sets presented in this work. At the DFT level we recommend def2-SV(P) basis sets to (usually) obtain (more than) qualitatively correct results and def2-TZVP bases to obtain results that are not too far from the DFT basis set limit, which is practically reached with the def2-QZVP bases. To achieve similar quality in HF calculations slightly larger polarization sets are needed, we thus recommend def2-SVP and def2-TZVPP bases for the respective purposes. In case of MP2 (and other post-HF methods) def2-SVP bases may be used only for explorative calculations, results that can be considered as quantitatively satisfactory are obtained with the def2-TZVPP bases. For the next step on the way to a consistent approximation to the MP2 basis set limit, def2-QZVPP bases may be used, which for MP2 calculations still show differences to the basis set limit of 0.01–0.1 eV $atom^{-1}$; similar errors may be expected for other post-HF methods.

Basis set recommendations for first-order properties—we have presented results for dipole moments as examples—are in general the same as for bond energies, furthermore for DFT calculations with def2-TZVPP bases (instead of def2-TZVP bases) results are close to the DFT basis set limit.

Also for optimization of structure parameters the above recommendations are valid. def-SV(P) results for DFT and def2-SVP bases usually are sufficient for qualitatively correct results, with def2-TZVP (def2-TZVPP) bases errors in bond lengths typically are smaller than 1 pm and that in bond angles smaller than 1° for the respective methods. At the MP2 level, structure optimizations with def2-SVP basis sets might be expected to yield qualitatively reasonable results only for p compounds; in any other case they only have rough exploratory character at most. If s or d elements are present, one better uses def2-TZVPP basis sets in connection with MP2 or any other post-HF method.

Availability

All bases mentioned above are available on the internet,[27] the def2-bases were submitted to the basis set library of the Environmental and Molecular Science Laboratory (EMSL, see ref. 2). The latter are also available as supplementary material, as well as the (Cartesian) coordinates of the molecules of the test set and all data (atomization energies and dipole moments) used for the statistical evaluations in the present work.

Acknowledgements

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26 The molecular test set contains the following 311 compounds; geometrydata (Cartesian coordinates) are available as supplementary material as well as all data used for the statistical evaluations in the present work BaF, BaF_2, BaH_2, BaO, BaS, Be_2F_4, Be_2H_4, Be_4, BeC_2H_6, $BeF_2O_2H_4$, BeH_2, BeS, $CaCl_2$, CaF_2, CaH_2, CsF, CsH, CsO, K_2S, K_3P, KBr, KCl, KF, KH, KI, Li_2, Li_2O, $Li_4C_4H_{12}$, Li_4Cl_4, Li_4H_4, Li_8, $LiBH_4$, LiCl, LiF, LiH, LiSLi, Mg_4, $MgCl_2$, MgF, MgF_2, MgH_2, Na_2O, Na_2S, Na_3N, Na_3P, NaCl, NaF, NaH, PLi_3, RbF, RbH, RbO, SrF, SrF_2, SrH_2, SrO, SrS, B_2H_6, $B_3N_3H_6$, B_4H_4, BF_3, BH_3, BH_3CO, BH_3NH_3, C_2H_2, C_2H_3N, C_2H_4, C_2H_6, C_4H_4, C_6H_6, CF_4, CH_2O, CH_2O_2, CH_3N, CH_3OH, CH_4, CO, CO_2, F_2, H_2, H_2CO_3, H_2O, H_2O_2, HCN, HF, HNC, HNO, HNO_2, HNO_3, N_2, N_2H_2, N_2H_4, N_4, NF_3, NH_3, NH_4F, OF_2, Al_2O_3, Al_2S_3, $AlCl_3$, AlF_3, AlH_3, AlN, CS_2, Cl_2, ClF, ClF_3, H_2SO_4, H_3PO_4, HCP, HCl, HSH, HSSH, P_2, PF_3, PF_5, PH_3, S_2, S_5, SF_2, SF_4, SF_6, $SiCl_4$, SiF_4, SiH_4, SiO_2, SiS_2, As_4, As_4S_4, $AsCl_3$, $AsCl_6^-$, AsH_3, Br_2, BrCl, BrO_4^-, GaCl, $GaCl_3$, GaF, GaF_2, GaH_3, GaO, $GeCl_4$, GeF_3, GeF_4, GeH4, GeO, GeO_2, HBr, $HCBr_3$, Se_8, SeH_2, SeO, SeO_2, I_2, ICl, IH, IO_4^-, InCl, $InCl_3$, InH, InH_3, InO, $SbCl_6^-$, SbF, SbF_3, SbH_3, SbO_2, SnF_3, SnH_4, SnO, SnO_2, TeF_3, TeH_2, TeO, TeO_2, XeF_2, XeF_4, $XeOF_4$, $BiCl_6^-$, BiF, BiF_3, BiH_3, BiO_2, PbF_3, PbH_4, PbO, PbO_2, TlCl, $TlCl_3$, TlH, TlH_3, TlO, $CoCl_3$, CoF_2, CoF_3, $Cr(CO)_6$, $CrCl_3$, CrF_3, CrO_3, Cu_2, Cu_2O, Cu_2S, CuCN, CuCl, CuF, CuH, $Fe(CO)_5$, FeF_2, FeF_3, FeO, ferrocene, MnF_2, MnO, MnO_2, MnO_3F, MnO_4^-, MnS, $Ni(CO)_4$, $NiCl_2$, NiF_2, NiF_3, NiO, NiS, $ScCl_3$, ScF_3, ScH_3, ScO, $Ti(CO)_4$, $TiCl_4$, TiF_3, TiF_4, TiH_4, TiO, TiO_2, TiS_2, VH_5, VO, VOF_3, $ZnCl_2$, ZnF_2, ZnH_2, $ZnMe_2$, Ag_2, AgCl, CdF_2, $CdMe_2$, $Mo(CO)_6$, MoF_3, MoH, MoO_2, MoO_3, NbF_3, NbO, NbO_2, NbO_2F, $Pd(CO)_4$, PdF, PdO_2, RhF, RhF_4, RhF_6, RhO, $Ru(CO)_5$, RuF, RuO, RuO_2, RuO_4, Tc_2O_7, TcO, TcO_3F, YF, YF_3, YO, ZrF, ZrF_3, ZrO, ZrO_2, Au_2, Au_3^-, AuCl, $AuCl_3$, HfF, HfF_3, HfO, HfO_2, Hg_2Cl_2, HgF_2, $HgMe_2$, IrF_6, $Os(CO)_5$, OsO_2, OsO_3, OsO_4, $OsOF_5$, $Pt(CO)_4$, PtO, PtO_2, ReH, ReO, ReO_2, ReO_3, ReO_3F, TaF, TaF_3, TaO_2F, $W(CO)_6$, WF_3, WH, WO, WO_2, WO_3.
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